

ALBERT BAIDEN-AMISSAH

Full-Stack, IoT & Embedded Systems Engineer | Builder

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PROFILE

Computer Science & Engineering student who ships across the full stack — firmware, embedded hardware, web backends, and AI integration — and builds in public. Track record of taking an idea from CAD/prototype to working system independently: a mechanical hygiene station, a fingerprint-secured vending controller, a solar LoRa sensor network, and a local-first AI assistant, among others. Looking for an environment that funds exploration in emerging tech (IoT, embedded AI, automation) the way a fast-moving product team explores new edges — in exchange for a builder who ships fast, documents everything, and turns early experiments into shippable product.

PRACTICAL EXPERIENCE

IT Hardware Engineering Intern | Ghana Grid Company Limited (GRIDCo) *Aug 2025 – Sep 2025*

- Performed diagnostic troubleshooting, component-level testing, and preventive maintenance across IT hardware and peripheral systems within a national power-transmission utility environment.
- Collaborated with cross-functional IT teams to maintain technical reliability and system uptime.

SELECTED PROJECTS

Sanitary Hub Pro — *Smart Sanitary Pad Vending Machine*

- Designed and built a microcontroller-driven vending system (ESP32) using keypad access and fingerprint verification for automated, secure product dispensing.
- Built an integrated web dashboard for real-time inventory and system telemetry monitoring.

Emergency Hygiene Station — *Designathon — Disaster-Relief Hardware*

- Designed a fully mechanical, portable hygiene station in Autodesk Fusion 360: foot-pedal taps, foldable basins, and dual water sources, engineered for zero-power deployment in relief settings.

FocusFlow — *IoT Desk Productivity System*

- Prototyped an ESP32-based desktop hardware device synced to a cloud database (Supabase) to track and visualize study/task workflows in real time, with Pomodoro and subject-tagging logic.

Alverri — *Local-First AI Personal Assistant*

- Authored the complete Software Design Document and built a FastAPI-backed assistant with voice interaction, Whisper speech recognition, and local LLM routing (Gemini/Ollama fallback) for offline-first use.

Automated Core Logging & Recovery Tracking System — *Offline-First Field Tool*

- Built an offline-first geological core logging system with barcode integration, inventory tracking, sensor input, and background sync as a Progressive Web App (React).

Builder's Journal / Creative OS — *Project Documentation Tool*

- Built a local-first React/Next.js app (IndexedDB) with a Gemini-powered Story Engine that converts raw dev logs into structured, publishable build narratives.

Additional projects (Morse Code wireless messenger, BeulahX logistics backend, F-CODE, StuddyHub, robotics builds) documented at <https://my-portfolio-ten-theta-68.vercel.app>.

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript/TypeScript, SQL, PHP, HTML/CSS
- Web & Backend:** React, Next.js, FastAPI, REST APIs, Supabase, MySQL/MariaDB/MongoDB, Vite, Bootstrap
- IoT & Embedded:** ESP32, Arduino, ESP-NOW, LoRa, sensors, servo/stepper motors, OLED/LCD displays, PlatformIO
- AI & Computer Vision:** Local LLM integration, Whisper speech recognition, OpenCV, AI assistant architecture
- Tools:** Git/GitHub, VS Code, Fusion 360, Arduino IDE, XAMPP, WSL, Kali Linux

LEADERSHIP & COMMUNITY

- Tutor, Aaenics (university robotics & technology club) — mentored members on hands-on embedded and software projects; delivered a talk on the gap between club-level and industry-level engineering work.
- Member, Google Developer Groups (GDG) — UMaT chapter.
- Co-founder (in progress), community build lab — planning a workshop-style maker space run through Aaenics/GDG.

EDUCATION

BSc Computer Science and Engineering — University of Mines and Technology (UMaT), Tarkwa, Ghana

Level 300 (Third Year)

St. Augustine's College — Completed 2023

RELEVANT COURSEWORK

Data Structures & Algorithms, Database Systems, Computer Networks, Embedded Systems, Software Engineering, Operating Systems, Information Security

COMPETITIONS & CERTIFICATIONS

- Autodesk Designathon — Emergency Hygiene Station (mechanical/CAD design)
- UMaT Hackathon
- Smart City Builders Challenge — traffic congestion prediction (Random Forest / Gradient Boosting)
- Basic Ethical Hacking Certificate